

# Kihyeon Joo

M.S./Ph.D. Integrated Student · Department of Electrical and Computer Engineering, Inha University

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## RESEARCH INTERESTS

Robotics (vision-language-action models, human preference learning) · Human-Robot Interaction · Ethical AI (educational robots for moral development) · Brain-Computer Interfaces (EEG emotion recognition, EEG foundation models)

## EDUCATION

**Inha University**, Incheon, South Korea Mar. 2024 – Present  
M.S./Ph.D. Integrated Program in Electrical and Computer Engineering · Advisor: Prof. Byung Hyung Kim

**Inha University**, Incheon, South Korea 2024  
B.S. in Information and Communication Engineering · Advisor: Prof. Yongwoo Lee

## PUBLICATIONS

### Domestic Journals (in Korean)

- [J2] **K. Joo**, C. Woo, and B. H. Kim, “Diffusion-Based Data Augmentation for Few-Channel EEG Emotion Recognition Using Class-Conditional DDPM,” *Journal of Biomedical Engineering Research*, vol. 47, no. 1, pp. 1–10, Feb. 2026. [\[link\]](#)
- [J1] **K. Joo** and B. H. Kim, “AEGIS: A mmWave-based Wearable System for Efficient Human Detection in Low-Visibility Rescue Environments,” *Journal of Korea Multimedia Society*, vol. 28, no. 12, pp. 1914–1929, Dec. 2025. [\[doi\]](#)

### Domestic Conferences (in Korean)

- [C1] J. Yuk, **K. Joo**, Y. Park, and B. H. Kim, “Contact-Pressure-Invariant PPG Signal Extraction Model,” in *Proc. Korea Computer Congress (KCC)*, pp. 1151–1153, Jun. 2023.

## RESEARCH EXPERIENCE

**Graduate Researcher**, Affective Artificial Intelligence Lab, Inha University Mar. 2024 – Present

## RESEARCH PROJECTS

Development of XR Content Agent Technology Based on Emotion and Affect Reasoning Apr. 2026 – Dec. 2028  
Institute of Information & Communications Technology Planning & Evaluation (IITP) · Lead Researcher

## HONORS & SCHOLARSHIPS

**Inha Vision Fellowship**, Inha University 2026 – Present

**AI Convergence Innovation Human Resources Development Scholarship**, IITP / MSIT 2024 – 2025

## TEACHING EXPERIENCE

**Teaching Assistant**, Inha University – Reinforcement Learning; AI Creative Design 2024

## SKILLS

|                    |                                                 |
|--------------------|-------------------------------------------------|
| <b>Programming</b> | Python, PyTorch, Hugging Face                   |
| <b>Robotics</b>    | ROS2, MoveIt2, Isaac Sim, Intel RealSense, UR5e |
| <b>BCI / EEG</b>   | MNE-Python, Braindecode, scikit-learn           |
| <b>Environment</b> | Ubuntu, CUDA, Conda                             |